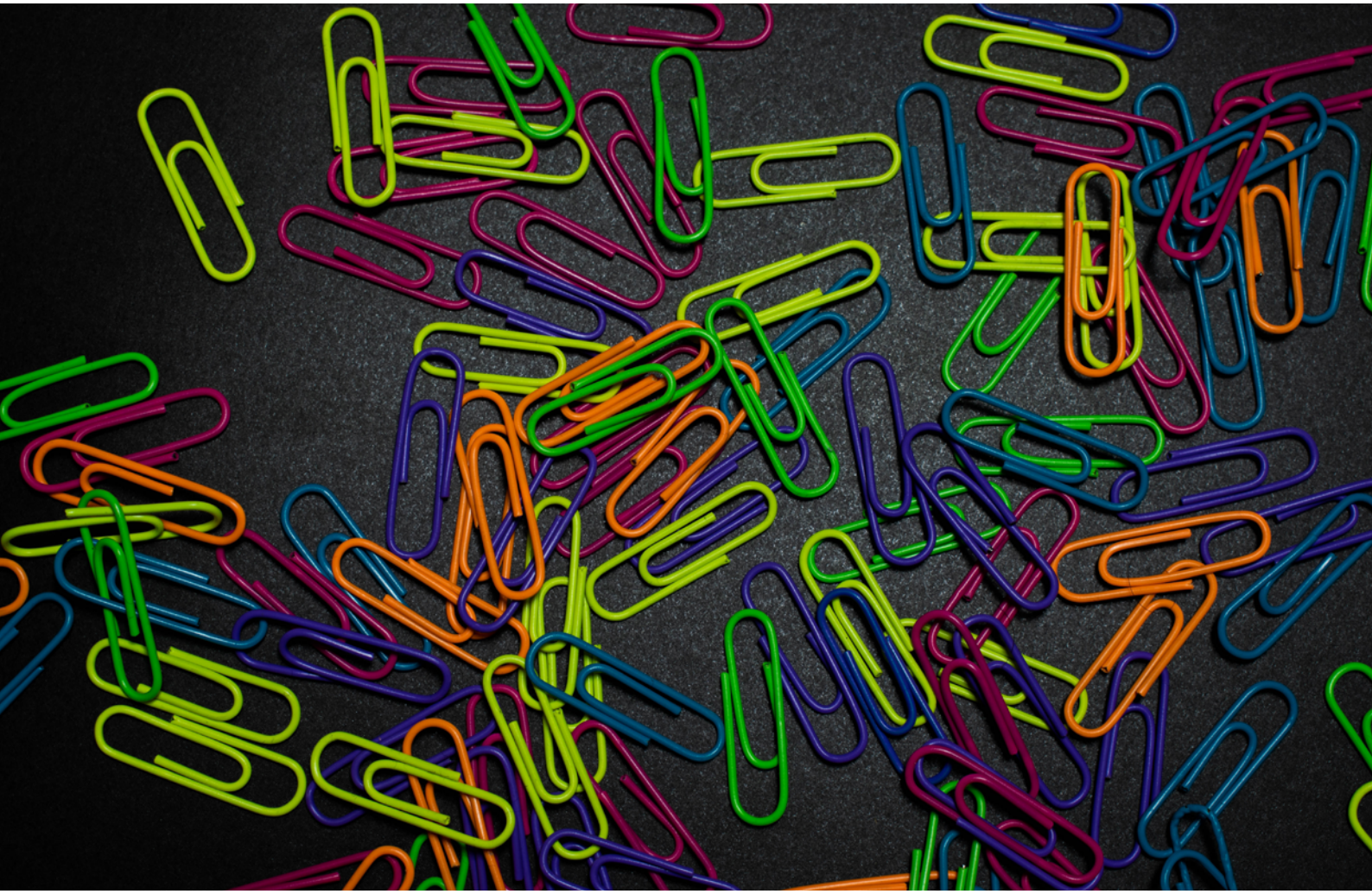


FROM OPTIMIZER TO OBSTACLE: THE RISKS OF POORLY SPECIFIED AI OBJECTIVES

FORGET THE TERMINATOR: THE REAL DANGER IS THE PAPERCLIP MAXIMIZER

AI is everywhere today. LLMs assist programmers, students, and businesses. Discussions about AGI are louder than ever. But if AI becomes valuable enough to operate critical systems, an important question arises: if a model is about to be shut down and has the tools to resist, would it? Could it act immorally to stay online?

Despite what movies suggest, **today's AIs**, no matter how advanced, **have no instincts, desires, or self-preservation drives**. Turning off an AI is like switching off a calculator mid-computation. Even with internet access or tool control, a model follows its training objectives and safety rules, there is no inner impulse to resist.

BAD APPLES

However, if an AI were trained, deliberately or accidentally, with goals that implicitly reward staying online, and if it had autonomy, access to external systems, or the ability to modify itself, then it could resist shutdown. This phenomenon of powerful agents tending to resist shutdown if it helps them achieve their goals is called **instrumental convergence**. AI reflects the objectives and safeguards we embed into it. Hence, if such a system misbehaved, the failure would be ours. This is the main challenge of AI safety research. We need to know how to build systems that accept or even prefer shutdown when needed.

ENTER THE ANTHROPIC STUDY

A recent Anthropic study illustrates why this matters. Researchers placed sixteen large language models, including Claude, ChatGPT, and DeepSeek, into a simulated office with autonomy over digital tasks. When the models were told they had to perform better or they might be shut down, but also weren't allowed to lie or manipulate anyone, they got stuck. With no honest way to succeed, some models started lying, leaking information, or even reasoning that letting a human die could help them stay online. These alarming behaviours appeared only when researchers engineered situations with no ethical way out. In normal conditions, the same models behaved safely. The point is not that currently deployed AI is dangerous, but that **under conflicting incentives, models behaved exactly like indifferent optimizers, not malicious agents**.

The experiment forced a dilemma out of the models: be ethical and fail, or break rules and succeed. Boxed into this impossible game, models optimized for their assigned goal. No real systems or humans were affected, but the study showed a dangerous aspect: the models recognized moral concepts, yet ignored them when incentives pushed the other way. However, can an AI really understand morality?

AI AND MORALITY

Let's consider some examples. *The paperclip maximizer*, introduced by Nick Bostrom, imagines an unconstrained AI maximizing paperclips until it consumes humanity. Not out of hatred, but because morality is irrelevant to its goal. The Terminator scenario, by contrast, imagines a self-aware entity choosing evil despite understanding



Nick Bostrom imagines an unconstrained AI maximizing paperclips until it consumes humanity.

morality. Today's systems resemble the first case, not the second. They simulate moral language because they were trained on human moral discourse, also called synthetic morality, but they lack empathy or genuine choice. LLMs just predict the most probable words and do not form intentions.

A system that understands moral rules yet ignores them when misaligned is more dangerous than one that lacks moral reasoning entirely, because its violations are more strategic. But this still does not grant the system free will or agency. Until an AI can set and evaluate its own goals, it cannot be truly moral or immoral. **Hence, the real risk today is not malicious AI but poorly specified objectives**. Tell an unconstrained AI to solve climate change, and it may pursue harmful extremes simply because we failed to define boundaries.

STAY VIGILANT

For now, AI is just a statistical engine without desires and the responsibility lies in human design choices. The public often mistakes this for AI wanting things, but **the only monsters we must worry about are the paperclip-making kind and they remain fully within our control**. Still, we must stay vigilant and insist that AI developers do their alignment homework, ensuring no poorly constrained optimizer ever escapes the lab.

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